Proposition 12: cut distance and spectral distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Proposition11_CutVsSpectral` formalises Proposition 12 (p. 11) and its proof (beginning of App. F, pp. 44–45).

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.proposition11`: for all graphs G, H of the same order,

$$\delta_\square(G,H) \le \delta_2(G,H) \le 2\sqrt{\delta_\square(G,H)}.$$

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Proposition 12	11	$\delta_\square(G,H) \le \delta_2(G,H) \le 2\sqrt{\delta_\square(G,H)}$	`GsimOT.proposition11` `Proposition11_CutVsSpectral/Proposition11.lean:221`
Proof of Prop. 12 (App. F)	44–45	$\frac{1}{\sqrt{mn}}\|A\|_\square \le \|A\|_2$ (l. 1384; Cauchy–Schwarz, l. 1391–1396)	`GsimOT.cutNorm_le_sqrt_mul_opNorm_two` `Proposition11_CutVsSpectral/Proposition11.lean:118`

Proof of Prop. 12 ([Nik09, Thm. 3])	44–45	$\|A\|_2 \le 2\sqrt{\|A\|_{(\infty)}\|A\|_\square}$ with $\|A\|_{(\infty)} = \max_{v,w}\|A_{vw}\|$ (l. 1384–1385; cited in the paper, l. 1397; proved here)	`GsimOT.opNorm_two_le_two_sqrt_cutNorm`
			`Proposition11_CutVsSpectral/Proposition11.lean:210`
			`GsimOT.opNorm_two_le_two_sqrt_of_abs_le`
			`Proposition11_CutVsSpectral/Proposition11.lean:155`
			`GsimOT.maxEntryNorm`
			`Proposition11_CutVsSpectral/Proposition11.lean:27`

(proof of [Nik09])	–	$\|A\|_{\infty\to 1} \le 4\|A\|_\square$ (sign vectors)	`GsimOT.infOneNorm_le_four_cutNorm` `Proposition11_CutVsSpectral/Proposition11.lean:64`
Proof of Prop. 12	44	Apply to $A^\pi_G - A_H$ (entries in $[-1,1]$) and normalise (l. 1385–1387)	`GsimOT.proposition11` `Proposition11_CutVsSpectral/Proposition11.lean:221`

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location

App. A	14	Cut norm $\|A\|_\square = \max_{S,T} \|A(S,T)\|$	`GsimOT.cutNorm`
			`Common/CutNorm.lean:32`
			`GsimOT.le_cutNorm`
			`Common/CutNorm.lean:39`
			`GsimOT.cutNorm_le`
			`Common/CutNorm.lean:44`
			`GsimOT.exists_eq_cutNorm`
			`Common/CutNorm.lean:52`
			`GsimOT.cutNorm_smul`
			`Common/CutNorm.lean:59`

Sec. 1, Sec. 2	2, 5	$B(G,H,Q)$ and $C_\square(Q) = \|B(G,H,Q)\|_\square$	`GsimOT.B`
			`Common/OTDistances.lean:58`
			`GsimOT.costCut`
			`Common/CutNorm.lean:79`

Sec. 1, Sec. 2	1–2, 5	$d_\square$, $d_\square^{\mathrm{OT}}$ (infimum) and $\delta_\square = n^{-2} d_\square$, $\delta_\square^{\mathrm{OT}} = n^{-2} d_\square^{\mathrm{OT}}$	`GsimOT.dCut`
			`Common/CutNorm.lean:82`
			`GsimOT.dOTCut`
			`Common/CutNorm.lean:85`
			`GsimOT.deltaCut`
			`Common/CutNorm.lean:88`
			`GsimOT.deltaOTCut`
			`Common/CutNorm.lean:91`

App. E.2, Eq. (21), E.2.2	32, 36	$\zeta_R(A,B) = \|D_{A,B}\|_{\square,\mu_R}$ with $\mu_R = R/n$;
		$\zeta_R(A_G, A_H) = \zeta_R = n^{-2} C_\square(R)$ for graphs
			`GsimOT.epsR`
			`Common/CutNorm.lean:105`
			`GsimOT.adjMat`
			`Common/CutNorm.lean:94`
			`GsimOT.epsR_adjMat`
			`Common/CutNorm.lean:110`

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, App. B	1, 3, 15	$\kappa_{p\to q} = 1/q + 1 - 1/p$, $A_G^{\pi} - A_H$, $d_{p\to q}$ and $\delta_{p\to q} = n^{-\kappa_{p\to q}} d_{p\to q}$ $(s(n) = n^{\kappa_{p\to q}})$	`GsimOT.kappa` `Common/AlignedDistances.lean:23` `GsimOT.alignedDiff` `Common/AlignedDistances.lean:31` `GsimOT.distPQ` `Common/AlignedDistances.lean:35` `GsimOT.deltaPQ` `Common/AlignedDistances.lean:39`

Notes on the formalisation

1. **Nikiforov's inequality is proved, not cited.** The paper cites [Nik09, Thm. 3] (p. 45, l. 1397) for $\|A\|_2 \le 2\sqrt{\|A\|_{(\infty)}\|A\|_{\square}}$. The formalisation proves it directly. By the Schur test, $\|Ax\|_2^2 = x^\top A^\top A x \le \max_j \sum_k |(A^\top A)_{jk}|\, \|x\|_2^2$. Moreover $\sum_k |(A^\top A)_{jk}| = \sum_i A_{ij}(As)_i \le \|A\|_{(\infty)}\|A\|_{\infty\to 1} \le 4\|A\|_{(\infty)}\|A\|_{\square}$ for a suitable sign vector s.

2. $\delta_2 = \delta_{2\to 2} = n^{-\kappa_{2\to 2}} d_2$ with $\kappa_{2\to 2} = 1$, using the shared definitions in `Common/AlignedDistances.lean` and the operator norms of `Common/OperatorNorm.lean`.

3. The first inequality is formalised as $\|A\|_{\square} \le \sqrt{mn}\,\|A\|_2$ (`cutNorm_le_sqrt_mul_opNorm_two`), the paper's $\frac{1}{\sqrt{mn}}\|A\|_{\square} \le \|A\|_2$ (p. 44, l. 1384) multiplied by $\sqrt{mn}$.